

HW 5.2.2

(a) $N(t) = 0$
 $\lambda = 60/\text{min}$

Since particles arrive independently & uniformly over time at a constant rate (60/min).

$$N(t) \sim \text{Poisson}(60t)$$

(b) $P(N(\frac{35}{60}) - N(\frac{35}{60}) = j | N(\frac{35}{60}) > 50)$

Since the Poisson process has independent increments, the t minutes is independent
 $N(t) \sim \text{Poisson}(60t)$

(c) Find distribution of particles by time $t \in [0, 6]$ min $\Rightarrow 100$
 $t \in [0, 6]$

$$P(N(t) = j | N(6) = 100) \sim \text{Binomial}(100, t/6)$$

(d) Find distribution of particles by time $t \in [0, 6]$
 $N(6) < 100$

$$P(N(t) = j | N(6) < 100) = \sum_{n=0}^{99} P(N(t) = j | N(6) = n) \cdot P(N(6) = n | N(6) < 100)$$

(e) Find distribution after 3 minutes

$$N(3) = 120$$

$$N(7) - N(3) \sim \text{Binomial}(120, 4/7)$$

(f) Find distribution after 2 minutes

$$N(2) > 120$$

$$\text{Let } N(7) = n > 120$$

$$P(N(7) - N(2) > k | N(7) > 120) = \sum_{n=121}^{\infty} P(N(7) - N(2) > k | N(7) = n) \cdot P(N(7) = n | N(7) > 120)$$

$$P(N(7) = n) \cdot P(N(7) = n | N(7) > 120)$$

$$\text{a mixture of Binomial}(n, 4/7)$$

(g) Suffices 120 particles hit over 8 minutes find!

$$P(2 \text{ particles in } [2, 4] \text{ 15 particles in } [7, 8] | N(8) = 120)$$

$$\text{Let } A = \text{Number in } [2, 4]$$

$$B = \text{Number in } [7, 8]$$

$$P(A=2, B=5 | N(8)=120) = \binom{120}{2, 5, 113} \left(\frac{2}{8}\right)^2 \left(\frac{5}{8}\right)^5 \left(\frac{5}{8}\right)^{113}$$

$$\text{Multinomial}(120, p_1 = 2/8, p_2 = 5/8, p_3 = 5/8)$$

5.2.3 -> Hm

Let T be the RV denoting time until relay.

$$T \sim \text{Exp}(\lambda)$$

$$CDF = F(t) = P(T \leq t) = 1 - e^{-\lambda t} \left\{ \int_0^t \lambda e^{-\lambda s} ds \right\}$$

In SIS model, T represent the RV time until

recovery of a given infected individual i .

$$P(\text{recovery within the next } \Delta t) = P(T \leq \Delta t) = 1 - e^{-\lambda \Delta t}$$

Hm 5.3.2

Given:

$$\lambda = 200 \text{ per 5 minutes} \Rightarrow \lambda = 40 \text{ per minute}$$

Let T_n be the waiting time till the n th particle hits the screen.

$$P(t_1 \leq T_n \leq t_2) = ?$$

$$T_n \sim \text{Exp}(\lambda)$$

$$T_n \sim \text{Gamma}(n, \lambda)$$

$$f_{T_n}(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}; t > 0$$

$$P(t_1 \leq T_n \leq t_2) = \int_{t_1}^{t_2} f_{T_n}(t) dt$$

$$\Rightarrow \int_{t_1}^{t_2} \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!} dt$$

HW 5-3-3

(a) $N(0) = 0$

$\lambda = 3/10 = 0.3$ Customer/min

$N(t) \sim \text{Poisson}(\lambda t)$

$N(t+s) - N(t) \sim \text{Poisson}(\lambda t)$

Customer arrives at independent occur at constant rate, and one-by-one.

(b) $P(T \leq 6)$:

$T \sim \text{Exp}(0.3)$

$P(T \leq 6) = 1 - e^{-(0.3)6} = 1 - e^{-1.8}$

(c) $E[T] = \frac{1}{\lambda} = \frac{1}{0.3} \approx 3.3 \text{ min}$

(d) $E[X_2 | 0 \leq X_1 \leq 8] = E[X_1] = \frac{1}{\lambda} = 3.33$

(e) $E[T_3 + T_4 + T_5] = 3 \cdot E(T) = 3 \cdot \frac{1}{0.3} = 10 \text{ min}$

(f) $P(2 \leq T_4 \leq 5) = F(5) - F(2) \quad \{T_4 \sim \text{Gamma}(4, 0.3)\}$
 $\Rightarrow \int_2^5 \frac{(0.3)^4 t^3 e^{-0.3t}}{3!} dt$

(g) Probability the 2nd Customer waits between 2 & 3 times?

$T_2 \leq 6$

$P(2 \leq T_2 \leq 3 | T_1 \leq 6)$

normalization

$T_2/T_1 \sim \text{Beta}(2, 3)$

$P(2 \leq T_2 \leq 3 | T_1 \leq 6) = P\left(\frac{2}{6} \leq \frac{T_2}{T_1} \leq \frac{3}{6}\right) = P\left(\frac{1}{3} \leq B \leq \frac{1}{2}\right)$

$\Rightarrow f_{\text{Beta}(2,3)}(0.5) - f_{\text{Beta}(2,3)}(1/3)$

5-33

(a) $S_5 \sim \text{Gamma}(5, 0.1)$

$E[S_5] = \frac{5}{\lambda}$

$\text{Var}(S_5) = \frac{5}{\lambda^2}$

$\sigma_{SS} = \sqrt{\frac{5}{(0.1)^2}} = \sqrt{5} = \sqrt{55.5} \approx 7.45 \text{ minutes}$

(b) Expected sample mean

$\text{mean} = E\left[\frac{1}{5} \sum_{i=1}^5 S_i\right], \text{Var}\left(\frac{1}{5} \sum_{i=1}^5 S_i\right)$

means

$E\left[\frac{1}{5} \sum_{i=1}^5 S_i\right] = \frac{1}{5} \sum_{i=1}^5 E[S_i] = \frac{1}{5} \sum_{i=1}^5 \frac{1}{\lambda} = \frac{1}{5 \times 0.1} = \frac{1}{0.5} = 2$

$\frac{1}{1.5} = \frac{5 \times 0.1}{2.5} = \frac{15}{15} = 1 \text{ minute}$

$\text{Var}\left(\frac{1}{5} \sum_{i=1}^5 S_i\right) = \frac{1}{25} \sum_{i=1}^5 \text{Var}(S_i) = \frac{1}{25} \sum_{i=1}^5 \frac{1}{\lambda^2} = \frac{1}{25 \cdot (0.1)^2} = \frac{1}{0.25} = 4$

$\Rightarrow \frac{15}{25(0.01)} = \frac{15}{0.25} = 60$

(c) Probability that 4th is sent before 8:50 a.m.

$\text{first} = 8:30 \text{ a.m.}$

$2^{\text{nd}} = 8:53 \text{ a.m.}$

$P(S_4 < 20 \text{ min})$

$S_4 \sim \text{Gamma}(4, \lambda = 0.3)$

$P(S_4 < 20) = F_{\text{Gamma}(4, 0.3)}(20)$

Homework 5.4.2

$$\lambda = \frac{6}{10} = 0.6 \text{ customer/min}$$

2) Joint Density of CDF

$$f(t_1, t_2, t_3, \dots, t_n) = \frac{\lambda^n}{t^n} \cdot \frac{6!}{30^6} \cdot 0 < t_1 < \dots < t_n < 30$$

Joint CDF

$$F(t_1, \dots, t_n) = P(t_1, \dots, t_n) = \frac{6!}{30^6} \cdot \lambda^n$$

3) Probability:

$$P(t_1 \in (4, 2], t_2 \in [4, 5], \dots, t_n \in [11, 20])$$

$$\Rightarrow \frac{6!}{30^6} \prod_{i=1}^n \Delta t_i \approx \frac{720}{30^6} \cdot (0.2)^6$$

4) Expected Sample

$$E\left[\frac{1}{n} \sum_{i=1}^n X_i - C\right]$$

t_n = time to failure from time 0, 30

$$E[t_n] = \frac{30}{\lambda}, \quad n=1, \dots, 6$$

$$E\left[\frac{1}{n} \sum_{i=1}^n S_i\right] = \frac{1}{n} \sum_{i=1}^n \frac{30}{\lambda} = \frac{30}{\lambda} \cdot \frac{1}{n} \sum_{i=1}^n 1 = \frac{30}{\lambda} \cdot \frac{6}{6} = 15$$

$$E\left[\frac{1}{n} \sum_{i=1}^n t_i - C\right] = 15 - C = 9 \text{ minutes}$$

5) 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30

$$f(t_1, \dots, t_n) = \frac{\lambda^n}{t^n} \cdot \frac{6!}{30^6} \cdot 0 < t_1 < \dots < t_n < 30$$

6) $E\left[\sum_{i=1}^n t_i\right]$

$$E[t_n] = \frac{30}{\lambda}, \quad n=1, \dots, 5 \Rightarrow \sum_{i=1}^n E[t_i] = \sum_{i=1}^n \frac{30}{\lambda}$$

$$\Rightarrow \frac{30}{\lambda} \cdot (1+2+3+4+5) = 15 \cdot 15 = 225 \text{ minutes}$$

5.4.2

$$P(s_1 \leq t_1, s_2 \leq t_2, s_3 \leq t_3 | N(t_3)=3) = \frac{n!}{t^n} \cdot \frac{6!}{30^6}$$

$$(0 < t_1 < t_2 < t_3 < 30) = \frac{f(t_1, t_2, t_3)}{f(s_1, s_2, s_3)} = \frac{3!}{t^3}$$